

On the retrieval of the specular reflection in GNSS carrier observations for ocean altimetry

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[1] This paper presents an altimetric method for ocean monitoring by remote sensing. It uses carrier observations of reflected GNSS signals. The method is illustrated in a simulation study and applied to a long term experiment yielding an ocean tide spectrum. The altimetric concept is based on residual observations of Doppler frequency. A linear relation between Doppler residuals f_0 and height departures ΔH from the surface level is derived. In contrast to existing phase-based methods which are constrained by smooth ocean conditions, the frequency-based retrieval here described holds good for rougher ocean conditions. Two retrievals of Doppler residuals are distinguished: Tracking Retrieval and Spectral Retrieval. A simulation study investigates the performance of Spectral Retrieval for a rough ocean surface with a noise-like sea level deviation $\xi(t)$. Simulation settings were adjusted to reflection events in coastal experiments with an elevation range of [5...15] deg. In this range Tracking Retrieval tolerates a surface standard deviation $\sigma_\xi < 5$ cm, whereas Spectral Retrieval tolerates $\sigma_\xi \leq 30$ cm. These limits correspond to significant wave heights of 20 cm for Tracking Retrieval and 1.2 m for Spectral Retrieval. The simulation results are confirmed by applying the altimetric method to the experimental data. The recovery of continuous phase tracks in experimental data is onerous and Tracking Retrieval only works for a period of smooth ocean conditions (162 events). By contrast, Spectral Retrieval yields altimetric estimates throughout the whole experiment (2607 events). The altimetric time series extends over more than 60 days and results in a tide spectrum that resolves diurnal (K1) and semidiurnal (M2, S2) constituents. The formal precision for these estimates lies in the decimeter range.

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1. Introduction

[2] Radar altimeters were launched on different satellite missions e.g. TOPEX/Poseidon and Jason to measure the sea level [Nerem *et al.*, 2010]. Sea level observations especially in coastal areas, where the use of radar altimeters is difficult, are of high interest. Existing models of geophysical factors e.g. Earth's gravity field or ocean currents are insufficient for local predictions of sea level. Additionally sea level changes in the 21st century will be accompanied by storm surges and extreme flooding events as indicated by simulations for

different coastal areas [Lowe and Gregory, 2005; McInnes *et al.*, 2003]. Such extreme events are a hazard especially for densely populated estuaries. Hydrologic studies focus on such areas but lack quantitative analyses of inundation events [Hung *et al.*, 2012]. The altimetric monitoring of inundation is a challenge for radio sounding techniques. The high reflectivity of water surfaces in L-band where GNSS signals are transmitted allows studies on GNSS reflections (GNSS-R) and research in adapted methods for inundation monitoring.

[3] Altimetric information in GNSS-R is provided by the delay between direct and reflected signals. Surface height can be estimated either from the amplitude waveform (code-delay-altimetry) or from carrier observations (carrier-phase-altimetry). A crucial difference is the precision of the delay, which corresponds to the signal path precision. Amplitude waveform precision is heavily determined by the code chip length among other factors. For example the C/A code with full public access has a chip length $\tau_c \approx 1\mu\text{s}$, that corresponds to a signal path length of 300 m.

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[4] Reliable concepts to estimate the surface height from amplitude waveform have been developed e.g. [Rius *et al.*, 2010]. However, altimetric precision is restricted for C/A code to the meter range. A greater precision was achieved using P code observations [Lowe *et al.*, 2002] but this code is restricted in access and has less signal power. As well as code-based retrieval, codeless altimetric retrieval from amplitude waveforms has been proposed in *Martin-Neira* [1993]. First altimetric results using this technique have been recently reported [Rius *et al.*, 2012]. The concept is based on a straight correlation of direct and reflected signals. In this case, a clean replica of the specific code is not required for correlation purposes. Regardless of the technique employed, a critical overlap of direct and reflected peaks in the amplitude waveform will occur if relative code delay goes below the limit $\tau_c = 1$ chip. Waveform retrievals are therefore dedicated to airborne or spaceborne setups where a necessary delay $\tau > \tau_c$ is given for most configurations of transmitter and receiver or at least a separate acquisition of direct and reflected signals is possible with up-looking and down-looking antennae respectively.

[5] In ground-based setups, the configuration is limited by the elevation angle and by the vertical distance between receiver and ocean surface, so for most locations the delay is restricted to $\tau < \tau_c$. In such a range of τ carrier phase observations are preferred. For experiments at short vertical distances $h < 50$ m (e.g. on piers or bridges) reflections occur close to nadir and the reflected signal can be separated from the direct one using a nadir-looking antenna. For these short distances standard geodetic receivers can be used for altimetry [Löfgren *et al.*, 2011]. Continuous phase tracking of the reflected signal is only possible for a calm sea [Belmonte Rivas and Martin-Neira, 2006]. In ground-based experiments at greater vertical distances $h > 50$ m (e.g. on a cliff or a mountain) slant reflections far from nadir were used. Adopted receiver solutions are then required [Treuhaf *et al.*, 2001; Fabra *et al.*, 2010; Semmling *et al.*, 2011].

[6] Reflections are also observed during Radio Occultation experiments with the receiver on a spaceborne orbit ($h \approx 400$ km) in a characteristic configuration to the transmitter. Reflection events occur during these experiments when the direct and reflected signals propagate with only a small delay $\tau < \tau_c$ at grazing elevation angles [Beyerle and Hocke, 2001]. Global coverage of these events has been reported [Beyerle *et al.*, 2002] but an altimetric method could only be applied for a case study on occultation data [Cardellach *et al.*, 2004].

[7] Carrier observations of ocean reflected signals in general are heavily affected by surface roughness. Continuous phase tracking in a ground-based setup is possible for a smooth ocean surface. Experiments at small incidence angles report a significant wave height below 10 cm [Belmonte Rivas and Martin-Neira, 2006]. But in particular for large vertical distances ($h > 50$ m) and especially for spaceborne setups, diffuse scattering plays a role. Simulations of the Glistening Zone described in Cardellach [2001] show that greater vertical distance increases the footprint size of the ocean reflection. The larger the footprint, the greater the number of diffuse scatterers which contribute to the reflected power. The retrieval of continuous phase observations fails if the diffuse part of reflected power exceeds the specular part.

This poses a particular problem for remote sensing applications in the retrieval of phase.

[8] An important idea in the following method is the use of residual observations that are synthesized from a signal path model and interferometric observations (explained in Section 2.4). Residual observations have a long modulation period, i.e. long coherent integration is possible to reduce noise. Doppler residuals are introduced here as integrated observations. Two retrievals of residual Doppler will be considered. The first type is a Tracking Retrieval where the residual Doppler is retrieved from the mean slope of the phase track. This type of retrieval has already been described in Semmling *et al.* [2011]. The second type is a Spectral Retrieval which uses a Fourier Transform of residual observation.

[9] This paper is structured in four sections. Section 1 gives an introduction to the altimetric use of interferometric carrier observations. In section 2 the altimetric method is described. A planar model and a ray tracing model of the reflection are considered. Section 3 presents a simulation study based on the planar model to clarify the method. Both retrievals, Tracking and Spectral, are considered and the effect of a noise-like roughness term $\xi(t)$ is analyzed. Section 4 focuses on the Godhavn Experiment. A first case study estimates sea level under smooth ocean conditions. Rough ocean conditions are addressed in a second case study. The retrievals are compared for both cases and for general application during the experiment. Finally, a tide spectrum is derived using a long time series based on Spectral Retrievals.

2. Altimetric Method

[10] This method is based on the interferometric path which is the difference between the reflected signal path L_r and the direct signal path L_d

$$L(t) = L_r(t) - L_d(t). \quad (1)$$

[11] This path is time-dependent. It changes during a reflection event. ‘Event’ in this context is the time series of interferometric observations recorded during the passage of a GNSS satellite. For event modeling sea levels $H = H_S + \Delta H$ are defined that are constant in time. These levels have an unknown departure ΔH from true specular sea level H_S which is estimated inversely. Two models for $L(t, \Delta H)$ are described in the following subsections.

2.1. Planar Model

[12] In planar approximation, a flat horizontal surface parallel to the local horizon at the receiver is assumed. The interferometric path is expressed thus:

$$\begin{aligned} L(t, \Delta H) &= 2[H_R - H_S - \Delta H] \sin E(t), \\ &= L(t, 0) + m(t) \Delta H \end{aligned} \quad (2)$$

where E is the elevation angle of the observed transmitter and the brackets in the first line enclose the vertical distance between the receiver height H_R and sea level $H_S + \Delta H$. Heights H_R, H_S are defined w.r.t. a common reference level, typically a reference ellipsoid. The equation’s second line

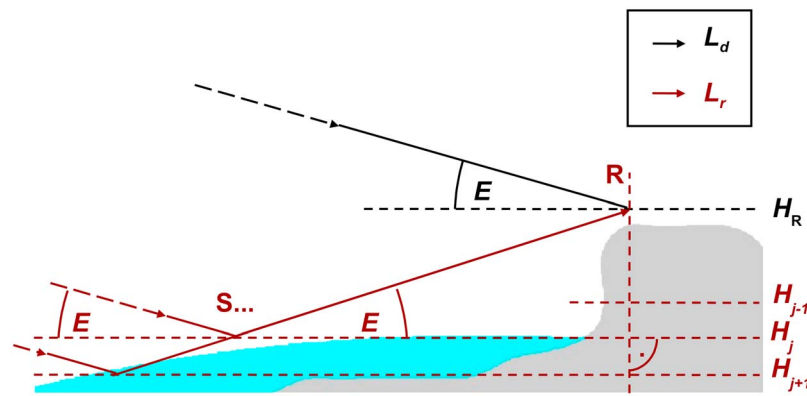


Figure 1. Scheme of the planar model. Planar surface levels are defined parallel to the horizon at receiver location R. The height levels H^j have an unknown departure ΔH^j from level H_S of the true specular point S. Earth's curvature and refractive bending are disregarded. The applicability of the planar model is therefore restricted.

expresses the linearity in departures ΔH . The planar approximation applies only for a short distance between receiver R and specular reflection S. For large distances Earth's curvature needs to be considered (see Figure 1).

[13] The transmitter is assumed at an infinite distance from the observed scene which means that incident rays at different heights are parallel with constant angles of elevation. Therefore the height dependence of E is disregarded and a linear coefficient $m(t)$ is introduced in the second line of equation (2). The planar model is sufficient to explain the principles of the altimetric concept. For altimetry in remote sensing where Earth's curvature and signal refraction need to be considered, a ray tracing model is introduced.

2.2. Ray Tracing Model

[14] A ray tracing tool was adapted to model specular reflections and to accommodate surface curvature and signal refraction. This was necessary due to the particular slant geometry of observations at the experimental site.

The model uses estimates of the transmitter trajectory and of the receiver position in an Earth-fixed reference frame (WGS-84). A specular reflection point is assumed to be located on a surface that is at rest in the Earth-fixed frame. A spherical surface is assumed. The surface refers to an Osculation sphere which is adjusted to WGS-84 ellipsoid. The sea level H can thus be considered an ellipsoidal height. Figure 2 shows spherical sea levels H^j referring to the adjusted Osculation.

[15] Obviously this model adapts better to Earth's curvature than the planar model (Figure 1). Direct path L_d and reflected path L_r are calculated along the rays shown in Figure 3.

[16] Note that path, in this context, includes tropospheric delays induced by the refractive medium. The difference $L = L_r - L_d$ yields the interferometric path length. An analytical form $L(t, \Delta H)$, as it was found for the planar model, cannot be given. The assumption of linearity can however be justified by a *Taylor Expansion* $L(\Delta H)$ around

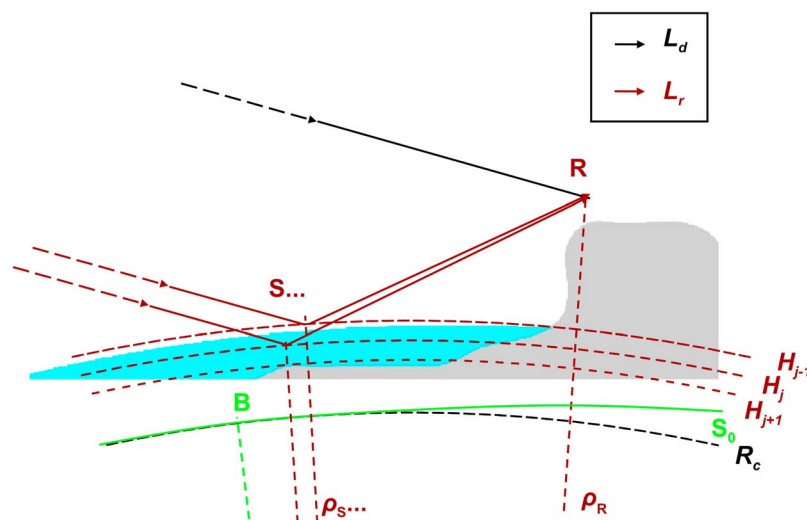


Figure 2. Scheme of surface levels H^j defined above the Osculation Sphere. The sphere fits the WGS-84 ellipsoid S_0 at base point B.

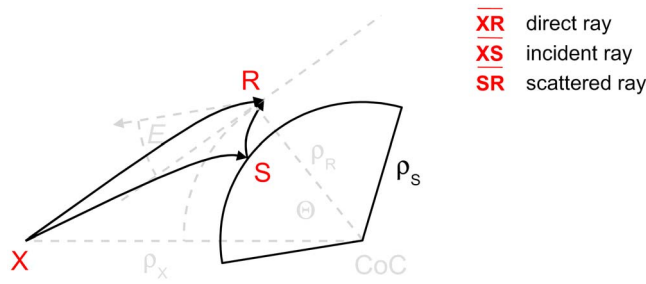


Figure 3. Ray propagation for the specular reflection. Direct, incident and scattered rays are shown. Their traces lie in one plane. A sphere with radius ρ_s defines the reflecting surface. The surface is concentric with the Osculation Sphere in the Centre of Curvature (CoC). Tropospheric refraction is indicated here by ray bending. It is modeled in concentric layers above the surface.

the true specular sea level H_s . This true level corresponds to a spherical shell with radius $\rho_s = R_c + H_s$ where R_c is the radius of the reference sphere (cf. Figure 2). Departures ΔH in the ray tracing model correspond to shells with $\rho = \rho_s + \Delta H$. The specular point and the corresponding path L for a given configuration of the receiver and transmitter are well defined for each shell. This means that, a function $L(\rho)$ exists. The maximum range of ocean undulations is $[-100 \dots 100 \text{ m}]$ (WGS-84), cf. the global range of Geoid undulations (EGM-96 model). Departures ΔH in the model can thus be restricted to this range. They are small compared to $\rho_s \approx 6371 \text{ km}$ and an expansion in $\rho = \rho_s$ is written

$$\begin{aligned} L(\Delta H) &= L(0) + L'(\rho_s)\Delta H + (1/2)L''(\rho_s)\Delta H^2 + o(\Delta H^2) \\ &\approx L(0) + m(\Delta H)|_{\Delta H=0}\Delta H \\ &\approx L(0) + m\Delta H. \end{aligned} \quad (3)$$

where the first and second derivatives w.r.t. ρ are indicated by one and two dashes respectively. Quadratic and higher terms can be disregarded. Calculations for a typical event, cf. parameters in equation (14) below, assuming maximum departures $|\Delta H| = 100 \text{ m}$ were conducted with the ray tracing model. The results at an intermediate elevation $E = 10^\circ$ are considered. In this limit of maximum departure the first term $L(0)$ is 237.18 m. The linear term is 34.33 m (14% of $L(0)$) and terms of an higher order contribute only 9 mm ($< 10^{-4} L(0)$). The second line therefore ends with the linear term, where the coefficient $m(\Delta H)$ is introduced. The indicated dependence on ΔH can also be disregarded. Calculations yield a difference between $m(0)$ and $m(100 \text{ m})$ of less than $10^{-4} m(0)$. Thus a linear relation as given in the third line can be assumed even for the ray tracing model. The path L is time dependent, of course. For simplicity t was left out in equation (3).

[17] Certain aspects like tropospheric refraction and the surface curvature are particularly relevant for the Godhavn Experiment. Other aspects which influence the GNSS signal path are disregarded e.g. ionospheric delays or the *Sagnac Effect* described in *Allan et al.* [1985]. The disregarded aspects affect similarly the direct and the reflected signal

path. The residual effect on the interferometric path is assumed to be negligible compared to the precision of retrievals described here.

2.3. Carrier Phase Observations

[18] The carrier phase can be used for altimetry based on the following equation of observation

$$\varphi_{\text{int}}(t) + \varphi(t_0) = \lambda^{-1}L(t, 0), \quad (4)$$

where λ is the wavelength of the L-band carrier, φ_{int} is the continuous interferometric phase in units of cycles and $L(t, 0)$ is the interferometric path without departure ΔH from specular sea level. The phase $\varphi(t_0)$ at the initial epoch t_0 is undetermined in the observations. Inserting the planar model equation (2) into this equation for a single epoch, unknowns $\varphi(t_0)$ and H_s remain. Least squares retrievals to estimate H_s have been proposed [*Martin-Neira et al.*, 2002; *Löfgren et al.*, 2011].

[19] To eliminate the unknown offset $\varphi(t_0)$ and to estimate H_s Doppler retrievals can be used. The general concept is briefly presented here before a detailed description of retrievals follows. By introducing the departure ΔH in equation (4) a residual phase φ_0 is defined

$$\varphi_{\text{int}}(t) + \varphi_0(t, \Delta H) = \lambda^{-1}L(t, \Delta H). \quad (5)$$

[20] The offset $\varphi(t_0)$ is assimilated by the residual. Subtracting equation (4) from equation (5) yields an explicit dependence of the residual on departures ΔH

$$\varphi_0(t, \Delta H) = \lambda^{-1}m(t)\Delta H + \varphi(t_0) \quad (6)$$

where the true specular path $L(t, 0)$ canceled and only the linear term with the model coefficient $m(t)$ remains after substituting the assumption $L(t, \Delta H) \approx L(t, 0) + m(t)\Delta H$. The time derivative of this equation removes $\varphi(t_0)$ and an equation for residual Doppler observations is obtained

$$\dot{\varphi}_0(t, \Delta H) = \lambda^{-1}\dot{m}(t)\Delta H \quad (7)$$

where the time derivative is indicated by the dot. Using a trial height $H = H_s + \Delta H$ with an a priori unknown departure ΔH the residual Doppler can be determined. The coefficient $\dot{m}(t)$ follows from calculations analytically using the planar model or numerically using the ray tracing model. The only unknown ΔH in equation (7) is determined and the true surface level H_s follows.

[21] Retrievals considered so far, least squares retrievals based on equation (4) as well as the Doppler retrieval in equation (7), require continuous phase data φ_{int} or φ_0 respectively. Ocean roughness however affects the continuity of GNSS-R phase observations. Discontinuities due to cycle errors occur when the reflected power fades. If these discontinuities cannot be corrected only phase fragments on a short time t are retrieved. Short fragments however restrict the altimetric use of phase data.

2.4. Residual Phasor Representation

[22] To understand fragmentation and to obtain a residual phase $\varphi_0(t, \Delta H)$ the early data level of the wrapped phase

argument ϕ_{int} is regarded. Correlation sums $I_{\text{int}}(t)$ (in-phase) and $Q_{\text{int}}(t)$ (quadrature-phase) are acquired by a dedicated receiver. They can be represented as a complex phasor observation $\gamma_{\text{int}}(t) = I_{\text{int}} + iQ_{\text{int}}$. The corresponding phase argument, $\phi_{\text{int}}(t) = \arg[\gamma_{\text{int}}(t)]$ in radian, is wrapped and contains discontinuities at the limits of the range interval $[0, 2\pi)$. The unwrap operation $\varphi_{\text{int}} = \mathcal{U}\{\phi_{\text{int}}\}$ corrects the phase by adding $\pm 2\pi$, when discontinuities greater than the tolerance of π occur. If SNR is lost due to power fading, cycle errors remain after unwrapping. A permanent loss of SNR makes it impossible to correct these errors manually and only fragments with a continuous phase $\varphi_{\text{int}}(t)$ are retrieved. Particularly for rapid phase changes cycle errors are critical.

[23] This problem is mitigated when the residual phase argument ϕ_0 with slow phase changes is introduced. The residual phase is created by counter-rotation of the observed phase using a phase model. In phasor notation the residual of rotation is written

$$\gamma_0(t) = \gamma(t)[\gamma_{\text{int}}(t)]^*, \quad (8)$$

where $\gamma(t) = \exp[-i2\pi \lambda^{-1} L(t, 0)]$ is the modeled phasor based on the true specular path $L(t, 0)$ and $[\gamma_{\text{int}}(t)]^*$ is the complex conjugate of the phasor observation. Provided that the model is correct, time dependent variations in the event vanish and the phase is stopped in its initial state at t_0 . An accumulation of observations is possible thus reducing noise and increasing precision as it has been previously reported for GNSS-R data by M. Caparrini et al. (PARFAIT: GNSS-R coastal altimetry, in *Workshop on Oceanography With GNSS Reflections*, arXiv:physics/0311052 [physics.ao-ph], 2003).

[24] Two options arise from residual phasor representation. On the one hand, unwrapping is significantly simplified that means the accumulated residual phase φ_0 is less fragmented than φ_{int} . On the other hand a spectral analysis of residual phasors γ_0 is possible. Both options are important for the retrieval of Doppler residuals.

2.5. Doppler Retrievals

[25] Two retrievals of a residual Doppler frequency are presented. For these residual frequencies the generalized symbol f_0 is used. Both retrievals are based on the residual phasor $\gamma_0(t, \Delta H)$ where in addition to time dependence t height departures ΔH are considered. The first way uses the mean slope of the residual phase track in the time domain (Tracking Retrieval)

$$\begin{aligned} f_0(\Delta H) &= \overline{\dot{\varphi}_0(\Delta H, t)} \\ \varphi_0(\Delta H, t) &= \frac{1}{2\pi} \mathcal{U}\{\arg[\gamma_0(\Delta H, t)]\}, \end{aligned} \quad (9)$$

where $\dot{\varphi}_0$ denotes the residual phase rate and the upper bar indicates the mean over all epochs t . The term $\arg[\gamma_0]$ denotes the wrapped phase argument in radian and $\mathcal{U}\{\}$ the unwrap operation. The mean must be restricted to epochs in continuous phase fragments without cycle errors. A coherence filter is used to find these epochs. The uncertainty δf_0 of the Tracking Retrieval depends on several parameters: phase noise, coherent integration time and number of regarded samples. For more details on the coherence filter and the uncertainty refer to Semmling [2011].

[26] A second retrieval of f_0 is based on the spectral peak of the residual phasor (Spectral Retrieval)

$$\begin{aligned} f_0(\Delta H) &= f[\hat{\Gamma}_0(\Delta H)] \\ \Gamma_0(\Delta H, f) &= \frac{1}{N} |\mathfrak{F}\{\gamma_0(\Delta H, t)\}|, \end{aligned} \quad (10)$$

where $\mathfrak{F}\{\}$ denotes the operator of the Discrete Fourier Transformation, N is the total number of samples and $\hat{\Gamma}_0$ is the peak in the Fourier amplitude. The critical unwrap operation needed for Tracking Retrieval is not needed here. The amplitude of the complex Fourier Transform has a peak at the position of the residual Doppler f_0 . The spectral resolution $1/T$ is important in determining the peak position. The duration of observations T determines the resolution, which equals the uncertainty $\delta f_0 = 1/T$ for the Spectral Retrieval. The sampling rate f_s is less important. For a constant duration T a higher rate f_s increases the total number of samples $N = T f_s$ and increases the spectral range $f_{\text{max}} - f_{\text{min}}$ but it has no effect on the uncertainty δf_0 .

[27] In general the Doppler in GNSS observations is not stationary and a Fourier analysis is not suitable. After counter-rotation in equation (8), however, the residual γ_0 is almost stationary and a distinct residual Doppler can be retrieved in the frequency domain.

2.6. Height Estimation

[28] Based on equation (7) a simple relation between Tracking Retrievals f_0 and height departures ΔH can be derived. Assuming f_0 is retrieved from a perfectly unwrapped phase Doppler residuals have the following relation to height departure

$$f_0(\Delta H) = \lambda^{-1} \overline{\dot{m}(t)} \Delta H \quad (11)$$

where dot and upper bar denote time derivative and temporal mean of the linear coefficient m respectively. Unfortunately a simple relation for Spectral Retrievals cannot be given. Postulating that both retrievals are equivalent for smooth surface conditions height inversion is generalized for both retrievals.

[29] For an idealized smooth ocean a single trial $H^j = H_S + \Delta H^j$ is sufficient. The path $L^j(t, \Delta H^j)$ is determined in forward modeling. Only the height H^j is known a priori, the true surface level H_S and the departure ΔH^j result from inversion. A residual phasor $\gamma_0^j(\Delta H^j)$ is constructed and a Doppler residual $f_0^j(\Delta H^j)$ is retrieved either by Tracking or by Spectral Retrieval. Recalling equation (11) an inverse function is found

$$H^j(f_0^j) = \overline{\lambda \dot{m}(t)}^{-1} f_0^j + H_S, \quad (12)$$

the equation can be solved for H_S if m is calculated using the model.

[30] For rough ocean conditions, equation (11) is insufficient as perfect unwrapping cannot be assumed; see Figure 4 in the following simulation study. In this case it is recommended to use multiple trials H^j to estimate H_S . If a correlation between f_0^j and H^j is detected in agreement with equation (12), an estimate H_S is found by a linear fit $H^j(f_0^j)$ for $f_0 = 0$. The fit uncertainty determines the

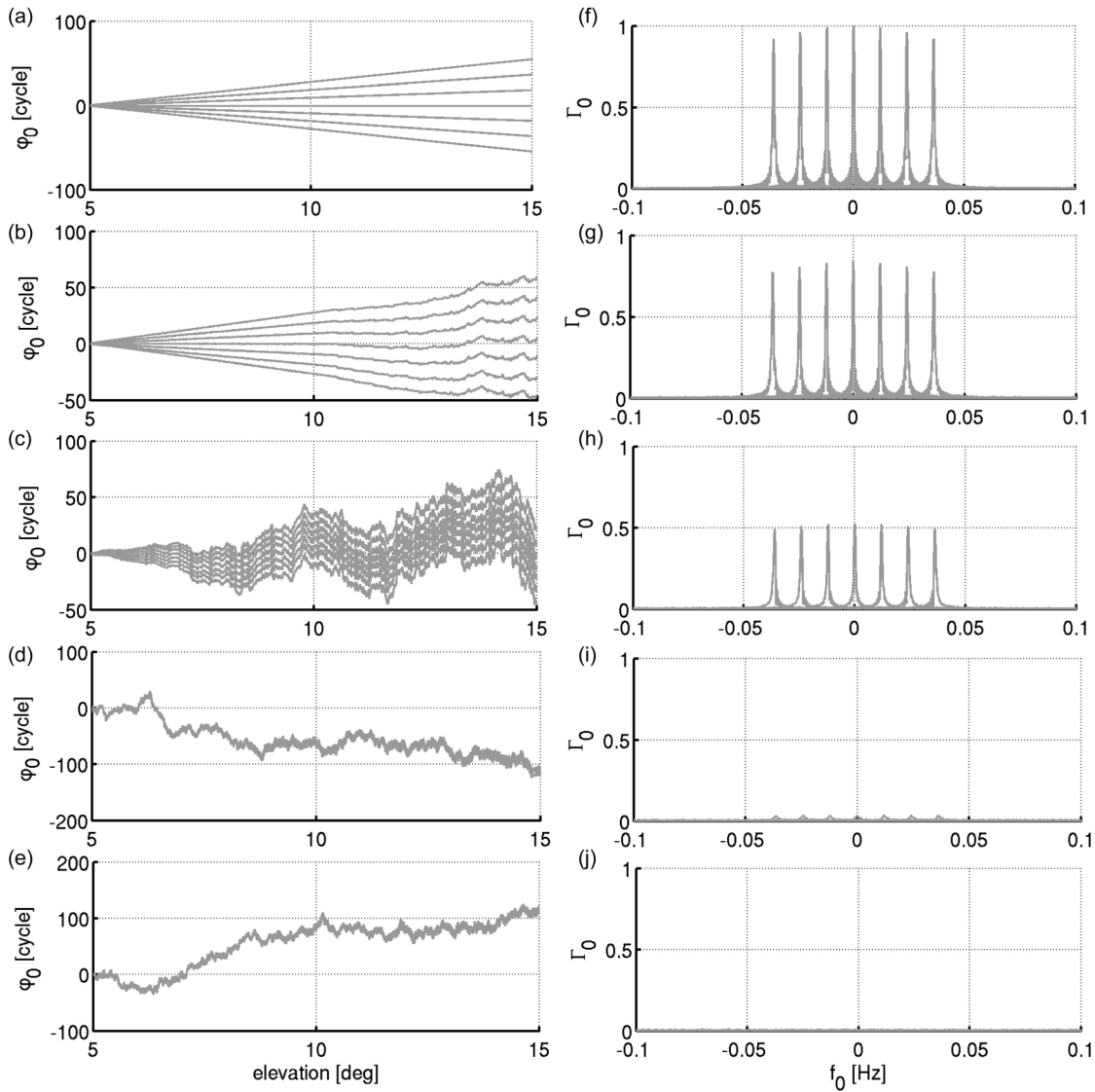


Figure 4. (a–i) Residual phase tracks versus elevation of the transmitter. (f–j) Fourier spectra versus residual Doppler. The retrievals are compared for realizations σ_ϵ from 0 cm (Figures 4a and 4f), 5 cm (Figures 4b and 4g), 10 cm (Figures 4c and 4h), 30 cm (Figures 4d and 4i), and 80 cm (Figures 4e and 4j). In each plot 7 tracks/spectra correspond to 7 predefined height levels.

quality of estimates. The formal precision of height estimates is given by

$$\delta H = \left| \overline{\lambda \dot{m}(t)}^{-1} \right| \delta f_0, \quad (13)$$

where δf_0 is the uncertainty of the Doppler residual.

3. Simulation Study

[31] The use of carrier phase observations is limited by roughness-induced signal fading, cf. Section 2.3. To evaluate the benefit of Doppler residuals and the corresponding retrievals, a simulation study follows that focuses on the roughness aspect. Based on the planar model in equation (2), specular reflection events are simulated under different roughness conditions.

[32] Ocean roughness has been widely studied in the context of GNSS scatterometry. The *Kirchhoff Approximation* is often used to model the ocean cross-section [Zavorotny and Voronovich, 2000; Cardellach and Rius, 2008]. A model of the reflected power, starting with ocean properties (reflectivity and distribution of slopes $\vec{s}(x, y)$ on the surface), is provided in this literature. Phase information, which is relevant for the presented method here, is not included in these models. Instead of using the distribution $\vec{s}(x, y)$ to describe ocean roughness, a deviation ξ from the undisturbed surface is examined. Only the time-dependent deviation $\xi(x, y, t) = \xi(t)$ is regarded to retain the specular path models. Momentary spacial variations that also contribute to phase noise are disregarded here. For $\xi(t)$ a normal Gaussian distribution is assumed with zero mean and a standard deviation σ_ξ . In other words, the simulated ocean

surface is perfectly smooth but the surface moves up and down according to time-dependent departures $\xi(t)$.

[33] The performance of the altimetric method is evaluated by perturbing the simulated events with $\xi(t)$. Results of the two retrievals are then compared. Thus different realizations with an increasing parameter σ_ξ are examined. The interferometric path of a simulated event in the planar model reads

$$\begin{aligned} L_{\text{sim}}(t, \xi) &= 2[H_R - H_\xi(t)] \sin E(t), \\ H_\xi(t) &= H_S + \xi(t) \end{aligned}$$

where H_R is the reference height of the receiver, H_S is the height of the undisturbed sea level and disturbance is introduced by the noise term $\xi(t)$.

[34] Using the same planar approximation the altimetric model reads

$$L^j(t, H^j) = 2[H_R - H^j] \sin E(t),$$

where H^j are the trial heights, cf. Figure 1. These heights and other parameters are set for the simulation according to conditions in the Godhavn Experiment (cf. Section 4):

$$\begin{aligned} H_R &= 700 \text{ m} \\ H_S &= 0 \text{ m} \\ H^j &= [-30, -20, \dots, 20, 30] \text{ m} \\ \sigma_\xi &= [0, 5, 10, 30, 80] \text{ cm} \\ E &= [5 \dots 15]^\circ \\ dE/dt &= 1^\circ/150 \text{ s} \\ \bar{m}_{L1} &= 758 \text{ m/Hz} \\ \bar{m}_{L2} &= 973 \text{ m/Hz} \end{aligned} \quad (14)$$

[35] For each realization of σ_ξ one event $L_{\text{sim}}(t)$ was simulated representing a single measurement. The elevation E changes in time, of course, but a constant rate dE/dt is assumed within the given range of the event. Linear coefficients $\bar{m}_{L1}$, $\bar{m}_{L2}$ are calculated for this event. The planar model allows the following analytical expression for the coefficient

$$\bar{m} = \frac{\lambda}{4\pi} \left[\cos E \frac{dE}{dt} \right]^{-1}, \quad (15)$$

where the elevation rate is given in radian per second and the upper bar indicates time average.

[36] Residuals are obtained by phase rotation $\gamma'_0 = \gamma_{\text{alt}}^j[\gamma_{\text{sim}}]^*$, where phasors are constructed for the altimetric model $\gamma^j(t, H) = \exp[-i 2\pi L^j(t, H)/\lambda]$ and the simulated event $\gamma_{\text{sim}}(t, \xi) = \exp[-i 2\pi L_{\text{sim}}(t, \xi)/\lambda]$. Only noise effects on the residual phase are analyzed here. Effects on amplitude are not considered. For departures $\Delta H^j = H^j - H_S$ of trial heights H^j from specular level, Doppler residuals f'_0 remain after phase rotation.

[37] Residuals f_0 are determined for both Tracking Retrieval and Spectral Retrieval in the following. Results are compared in Figure 4 where the noise parameter σ_ξ increases starting with smooth surface conditions.

[38] For $\sigma_\xi = 0$ phase tracks are continuous (Figure 4a) and the slope for each trial H^j can be derived. Roughness starts affecting the slope at $\sigma_\xi = 5$ cm (Figure 4b). Cycle errors occur in the unwrapped phase for elevation angles

above 10° . They appear when the variation between subsequent phase observations exceeds the tolerance π of the unwrap operation. The rapid increase in the number of cycle errors prevents their manual correction. Such a loss of continuous phase observations due to incoherent scattering is predicted by the *Fraunhofer Criterion* [Ulaby et al., 1982]

$$\hat{\sigma}_\xi = \frac{\lambda}{32 \sin \hat{E}}$$

where upper limits for the noise parameter $\hat{\sigma}_\xi$ and the elevation angle $\hat{E}$ are considered. The predicted elevation limit $\hat{E} = 7^\circ$ at $\sigma_\xi = 5$ cm agrees approximately with the shown realization. For realizations with $\sigma_\xi > 5$ cm the limit $\hat{E}$ lies below the simulated elevation range, i.e. only incoherent scattering is predicted. In agreement with those predictions, the phase track realizations with $\sigma_\xi = [10, 30, 80]$ cm lack any reliable slope information; see Figures 4c–4e.

[39] For Spectral Retrieval distinct peaks appear for each trial see Figures 4f–4i. Assuming smooth surface conditions ($\xi = 0$) a simple analytical expression can be found. The Fourier amplitude using specular sea level H_S is then written formally

$$\begin{aligned} \Gamma_0(H_S, f, \xi = 0) &= \frac{1}{N} |\mathfrak{F}\{\gamma_0(H_S, t, \xi = 0)\}| \\ &= \frac{1}{N} |\mathfrak{F}\{1\}| \\ &= \frac{1}{N} \delta(f), \end{aligned}$$

where $\mathfrak{F}\{\}$ is the discrete Fourier transform and N denotes the total number of samples in the event. In this case the residual phase vanishes completely and $\Gamma_0(f)$ corresponds to a *Dirac Distribution* with a singular peak at zero. Dependent on the departure ΔH the peak is distorted and shifted in the frequency domain (e.g., Figure 4f). As the noise parameter σ_ξ increases, so the peak amplitude decreases. Distinct peaks persist in the simulations up to $\sigma_\xi = 30$ cm (Figure 4i). The noise effect culminates in a random distribution of the residual phase argument in the range $[0, 2\pi)$. In this case the distinct peaks $\hat{\Gamma}_0$ vanish to the noise level; see Figure 4j. Limits for this effect are not predicted by the *Fraunhofer Criterion*.

[40] This comparison shows that Doppler residuals are retrieved spectrally for $\sigma_\xi \leq 30$ cm whereas the Tracking Retrieval is limited to $\sigma_\xi \leq 5$ cm. Consequences for the correlation between Doppler residuals and height departures are shown in Figure 5.

[41] For $\sigma_\xi = 0$ the correlation is perfect in agreement with equation (12). The curve crosses the origin with a vanishing residual Doppler at $\Delta H = 0$. For $\sigma_\xi = 5$ cm the correlation persists for both retrievals. However, a departure $\Delta H = 2.5$ m remains at $f_0 = 0$ after Tracking Retrieval.

[42] Thus an application limit of Tracking Retrieval in agreement with the *Fraunhofer Criterion* is found. For Tracking Retrievals with $\sigma_\xi > 5$ cm there is no correlation between f_0 and δH . By contrast Spectral Retrievals lose their correlation only for $\sigma_\xi > 30$ cm. A complete loss of correlation occurs at $\sigma_\xi = 80$ cm, a gradual increase of ΔH

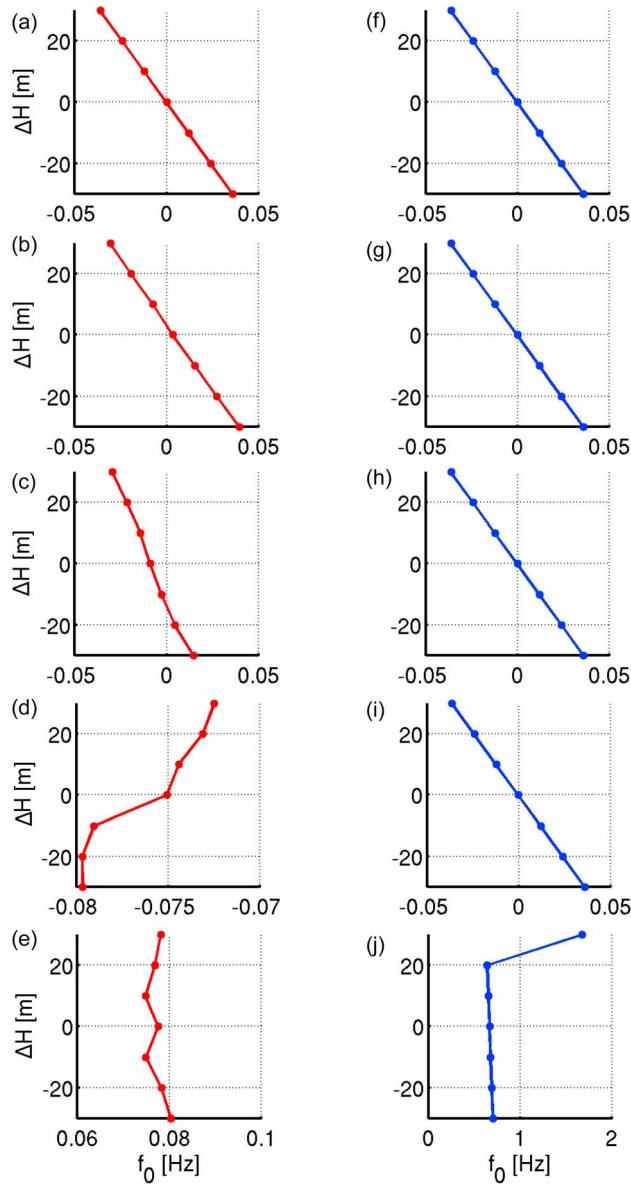


Figure 5. Correlation between residual Doppler and trial heights. The specular sea level H_S is fixed in the simulation and only the departure $\Delta H = H_S - H^j$ is considered. (a–e) Tracking Retrieval (red) and (f–j) Spectral Retrieval (blue) are compared. The noise parameter σ_ξ increases in 0 cm (Figures 5a and 5f), 5 cm (Figures 5b and 5g), 10 cm (Figures 5c and 5h), 30 cm (Figures 5d and 5i), and 80 cm (Figures 5e and 5j).

is not observed for Spectral Retrievals. Summarizing simulation results a better performance of Spectral Retrieval under rough ocean conditions is predicted. This prediction is verified in the following experiment.

4. Godhavn Experiment

[43] The altimetric concept using trial heights H^j to retrieve Doppler residuals has already been successfully applied to observations in the Godhavn Experiment

[Semmling et al., 2011]. Tracking Retrieval has been used in that study. During the extended period of a rough open-water surface tracks were disturbed and Tracking Retrieval could therefore not be applied. Altimetric results were only achieved during the sea-ice period. In this paper focus lies on the benefit related to Spectral Retrieval. Two case studies of individual events are described to illustrate the complexity of identifying phase tracks in the interferometric data. The relatively simpler retrieval using spectral peaks is presented by way of comparison. The long-term application of both retrievals is compared afterwards considering sea-ice and surface wind as roughness factors. A tidal study based on Spectral Retrievals over the whole period of experimental data follows at the end of this section.

[44] For precise analysis of experimental data the ray tracing model is used. The following parameters describe the conditions for altimetry at Godhavn:

$$\begin{aligned} E &= [5 \dots 15]^\circ \\ \alpha &= [+90 \dots 180 \dots -90]^\circ \\ H_R &= 691.2 \text{ m} \\ G_R &= 24 \text{ m} \\ H^j &= [G_R - 30 \text{ m}, G_R - 20 \text{ m}, \dots, G_R + 20 \text{ m}, G_R + 30 \text{ m}]. \end{aligned}$$

[45] Masks for the elevation E and azimuth α are applied. The receiver height H_R is determined by precise positioning. The Geoid undulation (EGM-96 model) at Godhavn provides a reference height G_R to define trial heights H^j . In principle more values out of the global range of Geoid undulations could be used, $[-100 \dots 100] \text{ m}$, to define trials H^j . This would, however, significantly increase computation time. More details about model and experiment can be found in Semmling [2011].

4.1. Case Study: Smooth Ocean Surface

[46] A reflection event of PRN 17 under smooth ocean conditions on 2009/01/18 is analyzed in this case study.

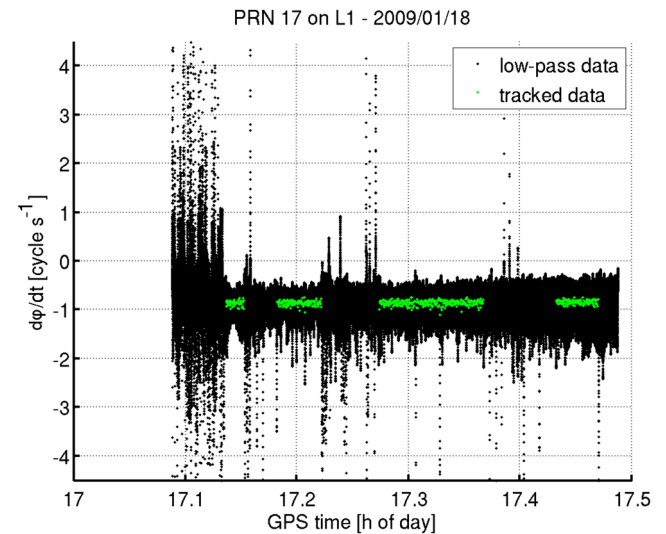


Figure 6. Phase rate versus hour of day for a smooth ocean surface. Data after low-pass filtering and data tracked by the dedicated algorithm are shown.

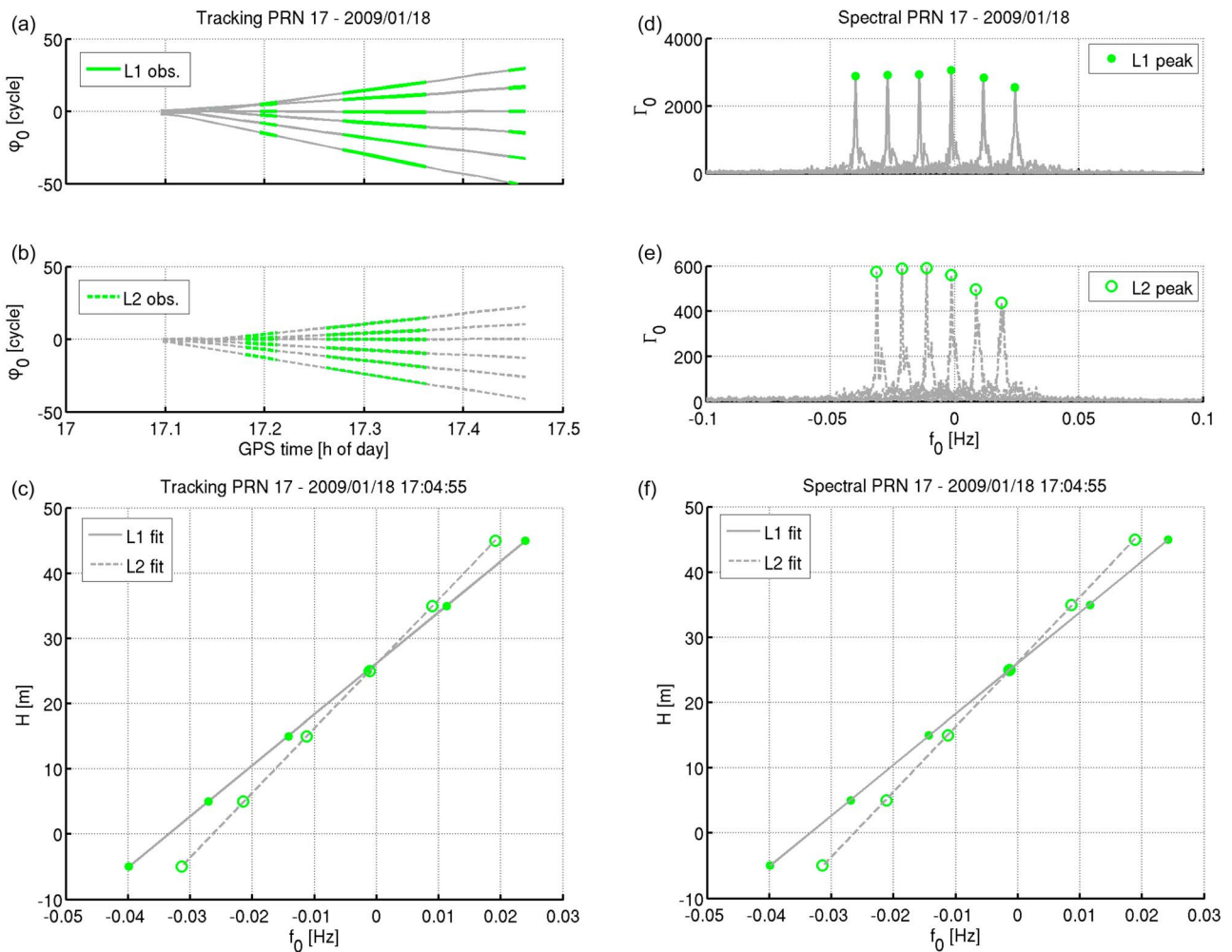


Figure 7. (a–c) Tracking Retrieval and (d–f) Spectral Retrieval, for a smooth ocean surface. Residual unwrapped phase versus hour of day are shown in Figures 7a and 7b. Six residuals at different trial heights are plotted. Tracks in green refer to coherent observations, and the gaps in between refer to incoherent parts. Coherence does not necessarily coincide for L1 and L2 observations. Spectra of the residual phasor with a zoom on the frequency axis for L1 and L2, respectively, are shown in Figures 7d and 7e. Spectral peaks marked in green indicate Doppler residuals at the six trial heights. Correlation of residual Doppler and trial height for Tracking and Spectral Retrieval, respectively, are shown in Figures 7c and 7f. Each panel contains two linear fits for height estimation on L1 and L2.

Much effort was spent at the beginning of data analysis to retrieve a continuous interferometric phase. Such a continuous track would be useful for carrier-phase-altimetry. Even for a smooth ocean surface, as in this case, there are parts in the event where the continuous phase is lost. An indicator is the phase rate. If large variations in the phase rate occur, a continuous phase track cannot be retrieved. A corresponding indicator for InSAR observations, the spatial phase gradient, is described in literature [Ghiglia and Pritt, 1998]. The phase rate in this case is shown in Figure 6.

[47] To reduce noise and to resolve phase rate characteristics a low-pass filter is applied to raw data. Bursts of the phase rate extending over 2 and more cycles per second occur in low-pass data. These bursts indicate fading parts

where a continuous phase cannot be retrieved. A dedicated phase tracking algorithm [Semmling and Helm, 2010] was used to find a continuous phase track (Figure 6 tracked data). The track is fragmented and not suitable for carrier-phase-altimetry.

[48] Doppler residuals are retrieved instead, first by Tracking Retrieval. Track fragments filtered by the tracking algorithm yield the residual phase in Figures 7a and 7b. Filtering is necessary, otherwise cycle errors in fading parts would bias the residual Doppler.

[49] Alternatively, the Spectral Retrieval can be applied. Then the residual Doppler is retrieved from spectral peaks in the frequency domain without a selection of tracks; see Figures 7d and 7e. The distinct peaks refer to different trial heights in the model. The peak distance agree with

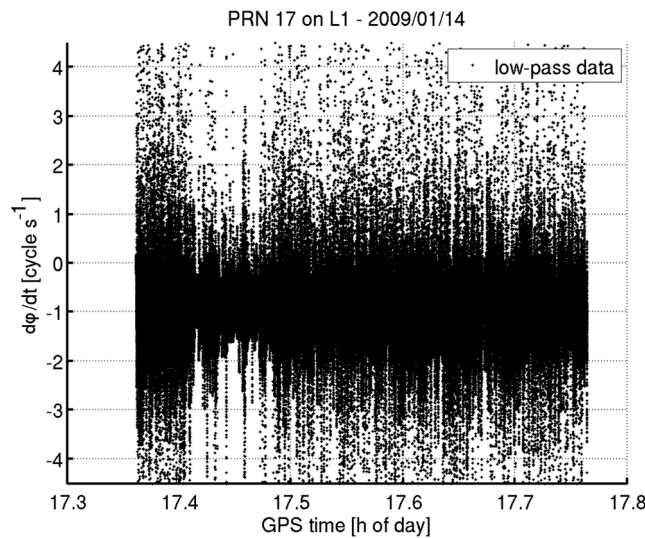


Figure 8. Phase rate versus hour of day for a rough ocean surface. Data after low-pass filtering is shown. No data could be tracked by the dedicated algorithm.

predictions of $\bar{m}\Delta H$ in equation (15). It is greater for L1 than for L2 peaks. The correlation between Doppler residuals and height levels for Tracking Retrieval and for Spectral Retrieval is shown in Figures 7c and 7f. In this case study under smooth ocean conditions, the results of both retrievals are similar. For Tracking Retrieval H_S is estimated in WGS-84, 26.2 m on L1 and 26.1 m on L2. For Spectral Retrieval it is estimated in WGS-84, 26.0 m on L1 and 26.3 m on L2. There is good agreement between both retrievals. Tidal effect corrections have yet to be applied. Unfortunately, an in-situ validation was not possible but an agreement with a tidal model is shown later. For purposes of comparison under different conditions of roughness, a second case study follows.

4.2. Case Study: Rough Ocean Surface

[50] A reflection event of PRN 17 under rough ocean conditions on 2009/01/14 is analyzed corresponding to the previous case study. Figure 8 shows the phase rate in this case.

[51] Again low-pass filtering is applied to reduce noise. The entire phase rate data spreads in a range of 3 and more cycles per second as opposed to isolated bursts in Figure 6 before. This spreading indicates the absence of any continuous phase information. As in the previous case tracking algorithm is employed but there is no tracked data in Figure 8 that can be used for Tracking Retrieval. The residual phase given in Figures 9a and 9b shows an irregular run.

[52] Spectral Retrieval is examined as an alternative. The absence of tracked data is not a fundamental problem as filtering is not required. The entire data set of the event is used to calculate the spectra. The results are shown in Figures 9d and 9e. Although the peak height is reduced compared with the previous case (Figures 7d and 7e), all spectra for L1 and L2 have distinct peaks. Despite loss of

continuous phase information, Doppler residuals can be retrieved.

[53] A correlation between Doppler residuals and trial heights is thus found but only for Spectral Retrievals (Figure 9f). Doppler residuals are also calculated based on Tracking Retrieval. The required filter of continuous tracks was omitted and the entire set of irregular phase data was used. Doppler residuals retrieved in this manner are not correlated with trial heights (Figure 9c). This means that by contrast with the previous case study where both retrievals gave similar results, in this case with a rough ocean surface only Spectral Retrieval provides sea level estimates H_S . Heights of 25.8 m for both L1 and L2 are obtained.

4.3. Long-Term Application

[54] The whole period of observations in the Godhavn Experiment lasted from 2008/11/20 to 2009/01/19. It comprises 2157 reflection events on L1 and 450 on L2. Only a small time period during sea-ice formation, the last 5 days, was used for Tracking Retrieval in the previous study [Semmling *et al.*, 2011]. There is a connection between the retrieval of continuous phase tracks and the smooth surface during sea-ice formation. In general, i.e. before sea-ice formation, the number of coherent observations forming continuous phase tracks is negligible. Figure 10 shows the distribution of coherent observations during the whole period.

[55] A typical event lasting 20 min with a 200 Hz sampling comprises 2.4×10^5 observations. Most events in Figure 10 have less than 0.5×10^5 coherent observations. This means that for the majority of events the fraction of continuous phase data ($<50\%$) is too small for the fading parts to be reconstructed. The lack of coherent observations explains the failure of Tracking Retrieval for large parts of the experiment.

[56] For Spectral Retrieval a selection of continuous tracks is not required so any event in Figure 10 would be suitable for Spectral Retrieval. The consistency of Spectral Retrievals is checked using a single trial $H^j = H_S$ given by an ocean surface model $H_S = G + z(t)$ which consists of a permanent offset G taken from the Geoid model EGM-96 and tidal variations $z(t)$ from the AODTM-5 model [Padman and Erofeeva, 2004]. Assuming a specular ocean reflection the retrieval is consistent if f_0 vanishes. Checking the consistency for all events, a distribution of Spectral Retrievals f_0 is plotted in Figure 11.

[57] The dominant blue color indicates consistent retrievals for most of the events. Outliers at frequencies $f_0 \neq 0$ arise from additional peaks in the spectrum due to other reflection sources, e.g. multipath coming from the snow covered surrounding of the setup; cf. also Fabra *et al.* [2011]. For specular ocean conditions the amplitude of the ocean peak exceeds the amplitude of other reflection peaks in the residual spectrum. When ocean roughness reduces the specular ocean reflection and the ocean peak vanishes, other peaks may persist and produce inconsistent outliers. In addition to inconsistent residuals f_0 , the coefficient $\bar{m}$ can be used to mask outliers. For the great distance between ocean surface and other dominant multipath

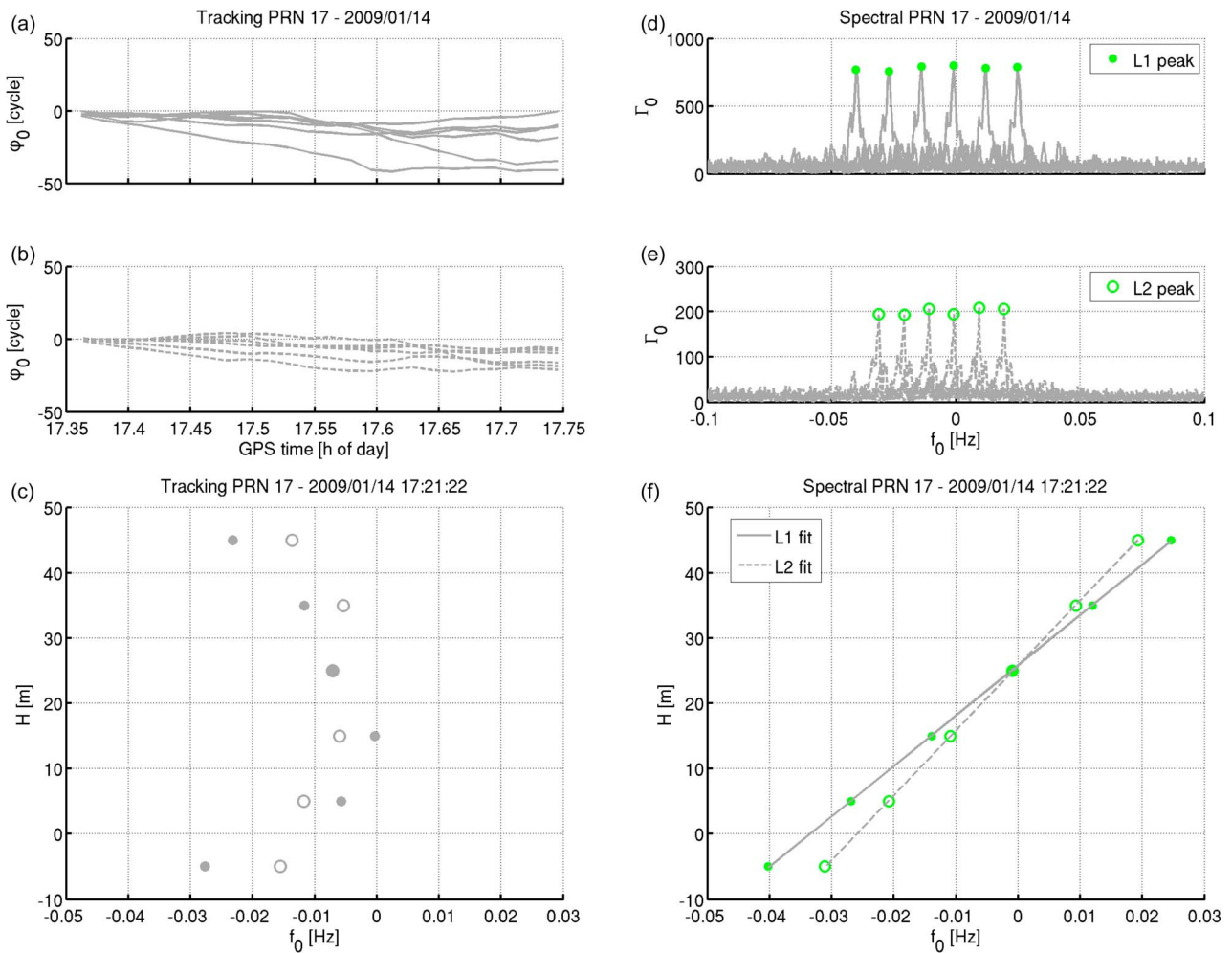


Figure 9. (a–c) Tracking Retrieval and (d–f) Spectral Retrieval for a rough ocean surface. Residual unwrapped phase versus hour of day are shown in Figures 9a and 9b. Six residuals at different trial heights are plotted for L1 and L2, respectively. No coherent observations were detected. The run of the phase in general is irregular. Spectra of the residual phasor with a zoom on the frequency axis for L1 and L2, respectively, are shown in Figures 9d and 9e. Spectral peaks marked in green indicate Doppler residuals at the six trial heights. Correlation of residual Doppler and trial height for Tracking and Spectral Retrieval, respectively, are shown in Figures 9c and 9f. Tracking Retrieval (Figure 9c) shows no correlation; a height estimation is therefore impossible. Spectral Retrieval (Figure 9f), instead, shows a correlation; the surface height is estimated using linear fits for L1 and L2, respectively.

sources the difference in $\bar{m}$ is significant and outliers can be masked easily.

[58] The occurrence of outliers is related to wind-induced ocean roughness. A comparison of surface wind velocity and outliers follows. Two clusters of outliers (Figure 11) coincide with high winds from southerly direction (Figure 12). One cluster occurs in the first third of December 2008, DoY [336...346], in particular for events recorded south-east of the receiver # [5...12]. A second cluster of outliers is found on 2009/01/14. On the same date, a low number of coherent observations (Figure 10) is reported. This cluster on 2009/01/14 contrasts significantly with events under smooth ocean conditions before and after that date. One explanation is the overall maximum in the south wind component

(Figure 12) which is reached on the same date and accompanied by a rough ocean surface.

4.4. Tidal Study With Spectral Retrievals

[59] Over the whole period from 2008/11/20 to 2009/01/19, the Spectral Retrieval yields a long time series of height estimates. The formal precision δH is calculated using equation (13). For a reflection event that typically lasts 20 min the formal precision is 50 cm on L1 and 70 cm on L2. Case studies discussed before have shown that the agreement of height estimates between L1 and L2 is better than indicated by formal precision. A single estimate is rather imprecise. The strength of Spectral Retrieval lies in the large number of events that are accessible.

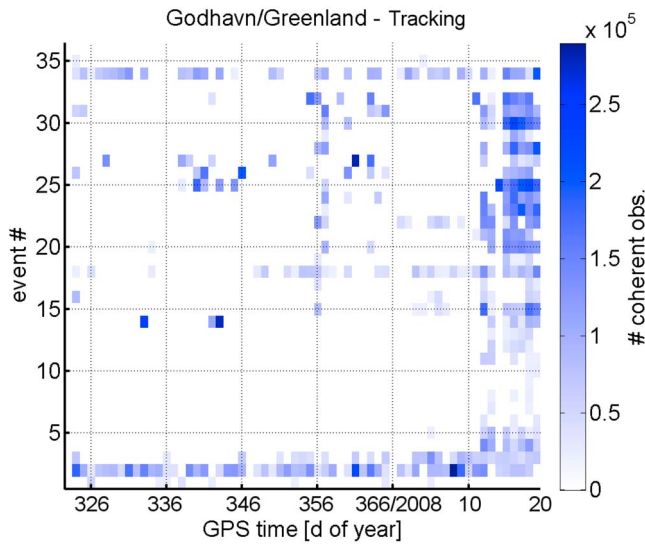


Figure 10. Number of coherent observations versus day of year and event number used for Tracking Retrieval. The whole observation period is considered. Events are sorted by azimuth. The event number is assigned clockwise from the east.

[60] The long time series is used to derive a tide spectrum for the ocean at the coast near Godhavn. The mean number of events per day, about 36 on L1 and 7.5 on L2, satisfies the minimum of 4 estimates per day that is required to resolve semidiurnal tides. The tide spectrum is derived from the time series of estimates H_S . The mean height $\overline{H_S(t)}$ is subtracted from the time series to remove the permanent offset of the ocean height. The difference $H_S(t) - \overline{H_S(t)}$ is interpolated linearly to a series $h_S(l)$ on an equidistant time axis t_l with a spacing $\Delta t = 15$ min, that is close to the mean spacing of

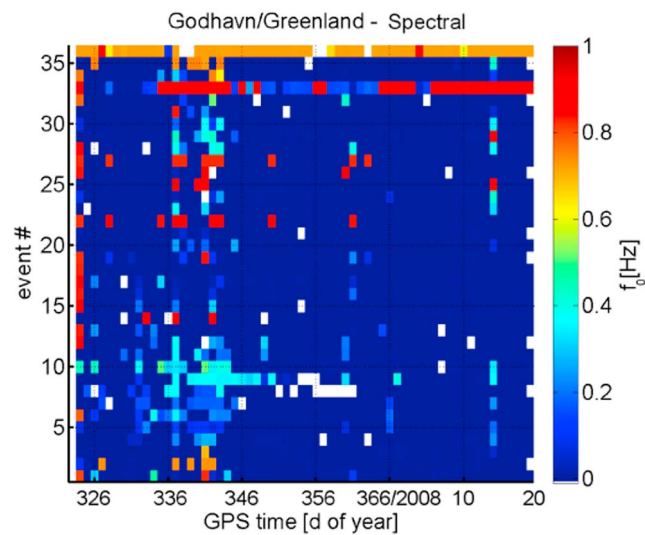


Figure 11. Residual Doppler versus day of year and event number determined by Spectral Retrieval. The absolute value of f_0 is plotted. The whole observation period is considered. Events are sorted by azimuth. The event number is assigned clockwise from the east.

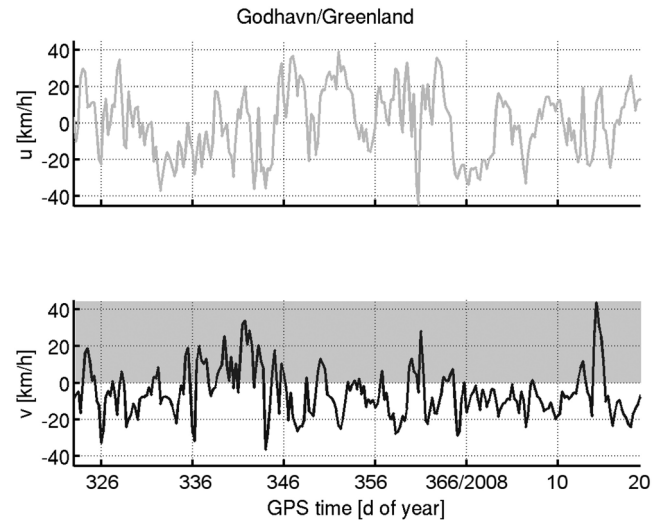


Figure 12. Horizontal surface wind velocity versus day of year. According to meteorological conventions, u component is positive for a west to east flow (west wind) and v component is positive for south to north flow (south wind shaded in gray). The data refer to the Godhavn area and were obtained from operational ECMWF analysis in postprocessing (all available measurements were assimilated). Model resolution is $0.5^\circ \times 0.5^\circ$ in longitude and latitude and 6 h in time.

20 min in the L1 records. The spectrum is then calculated using a discrete Fourier Transform

$$\eta(k) = \frac{1}{M} \sum_l^M h_S(l) e^{-i(l-1)(k-1)/M},$$

where the index l refers to the time axis $t_l = l\Delta t$ with M epochs in total and the index k refers to the frequency

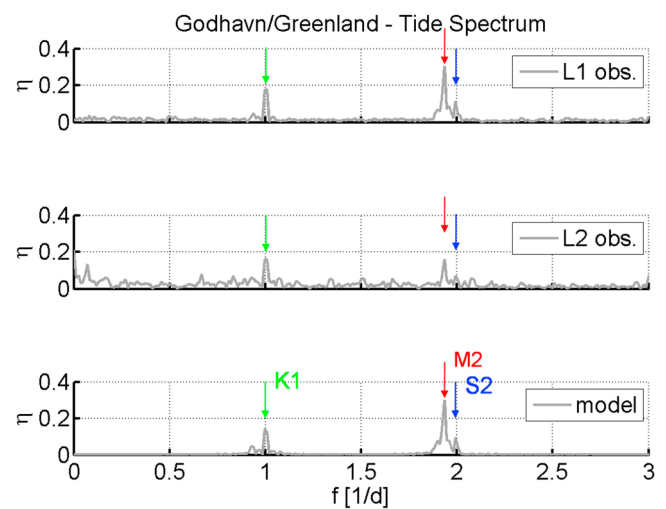


Figure 13. Tide spectrum with daily periodicity. Time series on L1 and L2 are considered. For comparison a modeled spectrum (AODTM-5 tide model) is added. The lunar semidiurnal constituents M2, the solar semidiurnal S2 and the lunisolar diurnal K1 are identified.

axis $f_k = k\Delta f$. Spectra are derived for L1, L2 observations and shown in Figure 13.

[61] The AODTM-5 model is plotted for validation.

5. Conclusion

[62] The presented method is based on an altimetric interpretation of Doppler residuals. Doppler residuals are retrieved based on residual phase tracks or on residual spectra. Trial heights in specular reflection models (a planar model or a ray tracing model) are used for the retrievals. If the reflection is specular, Doppler residuals are correlated with trial heights. A linear fit $H(f_0)$ for sea level estimation is justified.

[63] A simulation study examined the limits of Doppler retrievals due to roughness-induced noise $\xi(t)$ in a given geometric setting with elevation angles $[5...15]^\circ$. For Tracking Retrieval the unwrapping of a continuous phase fails for $\sigma_\xi > 5$ cm. A large number of cycle errors which cannot be corrected occurs above this limit and residual Doppler cannot subsequently be retrieved. Spectral Retrieval does not fail before $\sigma_\xi > 30$ cm when the peak amplitude in the residual spectrum vanishes to noise level. These limits correspond typically to significant wave heights of 20 cm (Tracking Retrieval) and 1.2 m (Spectral Retrieval). Spectral Retrieval is thus shown as the preferred method under ocean conditions.

[64] The experimental study confirms the simulation results. Inverse height estimation yields long-term altimetric results in particular using Spectral Retrieval. The method in general stands out due to noise reduction achieved by time integration of residuals. Time integration is important for both the mean slope in Tracking Retrieval as well as the Fourier Transformation in Spectral Retrieval. The advantage of Spectral Retrieval lies in an early reduction of noise. An additional transformation before time integration in the Fourier Transformation is not needed. Noise reduction in Tracking Retrieval however follows only after an unwrap operation. This operation is not well-defined and errors occur whenever the specular reflection power is fading. These errors affect the correlation of residual Doppler and trial heights. They lead to a failure of Tracking Retrieval when Spectral Retrieval still works.

[65] Precision of Spectral Retrievals is limited by integration time. For a typical 20 min reflection event in the Godhavn Experiment precision is in the decimeter range. Despite restricted precision, the reliability of this method supports further remote sensing studies. The application of this method for airborne experiments retrieving Doppler residuals for a receiver in motion is an upcoming challenge.

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